

Group A
[Answer all the questions]

1. Answer any FIVE

5x2=10

- a) Mention the key implementation that helped Java make it independent of platforms? What is its impact on performance in case of speed issues?
- b) What is the return type of a constructor? How can we call the constructor of the superclass from its subclass?
- c) "Strings are immutable" — what does that mean?
- d) Differentiate throw and throws.
- e) Write a block of code to print all the elements of an array containing string type members using the for-each version of loops.
- f) Write down the purpose of the operator ~ and << in Java.
- g) Clarify the relationship between class and object. Mention the operator that plays a key role in this relationship?
- h) "Redundant parentheses may degrade the performance of your program" — justify this statement.

2. Answer any FOUR

4x5=20

- a) Is the given code below alright? If not then correct the errors and write the output of the code. 5

```
int sum = 0;
int nums[ ] [ ] = new int[3][5];
for (int i = 0; i < 3; i++)
    for (int j = 0; j < 5; j++)
        nums[i][j] = (i+1) * (j+1);
for (int x | : nums) {
    for (int y : x) {
        System.out.println("Value is : " + y);
        sum += y;
    }
}
System.out.println("Summation : " + sum);
```

- b) Explain why main() is declared as public and static in a Java Program. Is the following code correct? Explain. 5

```
import java.lang.Math;

// inside class
double test = Math.PI * 5;
```

- c) Define abstract class. What are the criteria for one class to extend an abstract class? Present a scenario that justifies the use of an abstract class. 2+1+2
- d) Write a method maintaining the given criteria.

- 1. Suppose the method is given an array of integer numbers. This array will be traversed and each number will be inverted. Inverted numbers should be printed in the method.
- 2. If any number is found zero the program will throw an ArithmeticException and print that "A Zero Is Found!!!"
- 3. Whether any exception occurs or not the method will end printing "Array Traversing Stopped!!!"

- e) Explain the reason for Java's not supporting multiple inheritances. How can you implement this concept in Java? What is the name of the thread that begins immediately when a Java program starts?

2+2+1

- f) Create a class **Account** with two overloaded constructors. The first constructor is used for initializing the name of the account holder, the account number, and the initial amount in the account. The second constructor is used for initializing the name of the account holder, the account number, addresses, the type of account, and the current balance. The **Account** class has methods **deposit()**, **withdraw()** and **getBalance()**. Make the necessary assumption for data members and return types of the methods. Create objects of **Account** class and use them.

5

3. Answer any TWO

2x10=20

- a) 1. What does it mean when a class is declared **final**? What happens if an abstract class is declared **final**?
2. What type of class can be declared as **private**? What is the issue with declaring a class as **private**?
3. Define singleton class.
4. At what point is the **Bar** object, created on line 6, eligible for garbage collection?

2+2+2+

4

```
1. class Bar{
2. }
3.
4. public class Test{
5. Bar doBar(){
6. Bar b = new Bar();
7. return b;
8. }
9. public static void main(String args[]){
10. Test t = new Test();
11. Bar newBar = t.doBar();
12. system.out.println("newBar");
13. newBar = new Bar();
14. system.out.println("finishing");
15. }
16. }
```

- b) Check the following code carefully.

2+2+2+

2+2

```
public class Person{
    public void work(){
        system.out.println("Working");
    }
    public void sleep(){
        system.out.println("Sleeping");
    }
}

public class Teacher extends Person{
    public void work(){
        system.out.println("Teaching");
    }
    public void work(String course){
        system.out.println("Teaching courses");
    }
}
```

Now write the output of the following codes with the explanation to your answer. If there is an error, specify it with reason.

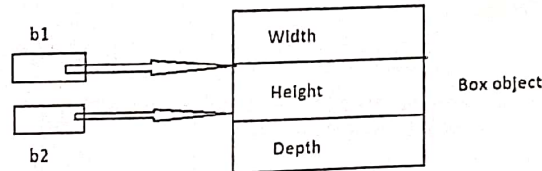
- i. `Person p = new Person();`
ii. `p.work();`
iii. `p.sleep();`
iv. `Teacher t = new Teacher();`
v. `t.work("OOP");`
c) 1. What is the key advantage of using multithreading? How can multiple threads be created in Java?
2. How is synchronization accomplished in a multithreading environment?

5+5

Group B
[Answer all the questions]

5x2=10

4. Answer any FIVE
- a) Draw a block diagram to illustrate the relationships among JVM, JRE, and JDK.
- b) How can you prevent a method from being overridden?
- c)



Design the class named Box and write suitable instructions relevant to the image.

- d) try{
.....
} catch (ArrayIndexOutOfBoundsException ex1){ }
catch (Exception ex2){ }
catch (ArithmeticException ex3){ }

What type of issue may arise in this try-catch block? Why?

- e) How is a constant declared in Java?
- f) If both the access specifiers "default" and "public" make members and methods of a class accessible from outside the class then what is the difference between them?
- g) `int a = 13;
byte b;
b = a;`

What type of error may this code create and why? Rewrite it to solve the issue.

- h) Suppose packages A and B both have classes named C. Now, what will happen if both the packages are imported using the * form?

4x5=20

5. Answer any FOUR

- a) `public class MyClass {
 int a = 3, b;
 void meth(int x) {
 System.out.println("x = " + x + "a = " + a + "b = " + b);
 }
 static {
 System.out.println("Static block initialized.");
 b = a * 4;
 }
 public static void main(String args[]){
 meth(42);
 }
}`

Write the output of the given code. If you think it has bug(s) then rewrite it, fixing the bug and mentioning the causes.

- b) What is method overriding? What happens if we change the arguments of the overriding method? How do you refer to the superclass version of the overridden method in the sub-class? Can we override a private method?

2+1+1+
1

c) What happens when a superclass variable refers to a subclass object? Explain the concept with suitable code blocks.

1+4

d) Define the Invoice class for the following program.

5

```
class Main{
    public static void main (String[] args) {
        Invoice inv = new Invoice("2034", 2, 325.50);
        System.out.println(inv.getPartNo() + " " + inv.getQuantity() + " "
            + inv.printPerItem());
    }
}
```

e) What is the output of the program and give the logic behind your answer.

```
class A{
    int i;
    void display(){
        System.out.println(i);
    }
}
class B extends A{
    int j;
    void display(){
        System.out.println(j);
    }
}
class Test{
    public static void main(String args[]){
        B obj = new B();    obj.i = 1;    obj.j = 2;
        obj.display();
    }
}
```

f) Write down the purposes of the keywords and methods given below;
i. synchronized ii. protected iii. finally iv. notify() v. static

6. Answer any TWO

2x10=20

a) 1. Write a Generic method *minimum* that finds the minimum for the following three cases.

5+5

```
public static void main(String args[]){
    System.out.println("Min: " + minimum (-3, 40, 15));
    System.out.println("Min: " + minimum (16.6, 8.8, 7.7));
    System.out.println("Min: " + minimum ("mango", "guava",
        "orange"));
}
```

2. Define deadlock. Why is inter-thread communication necessary? What are the different possible states of a thread.

b) 1. What is Type Wrapper in Java?

1+2+2+

2. Define boxing and unboxing with examples.

2+3

3. Write statements to show autoboxing and autounboxing.

4. What is Generics? What flexibility it provides while programming in Java.

5. wait(). notify() and notifyAll() these three methods are available in every class. Why?

c) 1. What is the signature of a method? Define method overloading. Write a sample code that exemplifies method overloading.

5+5

2. Write down the output of the following program:

```
static void procA(){
    try{
        System.out.println("inside procA");
        return; }finally{
        System.out.println("ProcA's finally");
    }
}
static void procB(){
    try{
        System.out.println("inside procB");
        throw new RuntimeException("Demo");
    }finally{
        System.out.println("ProcB's finally");
    }
}
public static void main(String[] args){
    procA();
    try{
        procB();
    }catch(Exception e){
        System.out.println("Exception handled");
    }
}
```